

Table of Contents

- 1. BFS (Breadth-First Search)
- 2. DFS (Depth-First Search)
- 3. Dijkstra's Algorithm (using Min-Heap / Priority Queue)
- 4. Bellman-Ford Algorithm
- 5. Floyd-Warshall Algorithm
- 6. Kruskal's Algorithm (MST)
- 7. Prim's Algorithm (using Min-Heap / Priority Queue)
- 8. Topological Sort (Kahn's Algorithm / DFS-based)
- 9. Union-Find (Disjoint Set Union - DSU)
- 10. Tarjan's Algorithm (SCC - Strongly Connected Components)
- 11. Kosaraju's Algorithm (SCC)
- 12. Johnson's Algorithm (All-Pairs Shortest Path for Sparse Graphs)

time complexity of all graph algos

Here's a list of common graph algorithms with their time complexities:

1. BFS (Breadth-First Search)

- Time Complexity: $O(V + E)$

2. DFS (Depth-First Search)

- Time Complexity: $O(V + E)$

3. Dijkstra's Algorithm (using Min-Heap / Priority Queue)

- Time Complexity: $O((V + E) \log V)$
-

4. Bellman-Ford Algorithm

- Time Complexity: $O(V * E)$
-

5. Floyd-Warshall Algorithm

- Time Complexity: $O(V^3)$
-

6. Kruskal's Algorithm (MST)

- Time Complexity: $O(E \log E)$ or $O(E \log V)$
-

7. Prim's Algorithm (using Min-Heap / Priority Queue)

- Time Complexity: $O((V + E) \log V)$
-

8. Topological Sort (Kahn's Algorithm / DFS-based)

- Time Complexity: $O(V + E)$
-

9. Union-Find (Disjoint Set Union - DSU)

- Time Complexity: $O(\alpha(V))$ per operation (α = inverse Ackermann, very slow-growing)
-

10. Tarjan's Algorithm (SCC - Strongly Connected Components)

- Time Complexity: $O(V + E)$
-

11. Kosaraju's Algorithm (SCC)

- Time Complexity: $O(V + E)$
-

12. Johnson's Algorithm (All-Pairs Shortest Path for Sparse Graphs)

- Time Complexity: $O(V^2 \log V + VE)$
-

Let me know if you want this categorized by weighted/unweighted, directed/undirected, or dense/sparse graphs.